

Performance Analysis of Deep Learning models in Deepfake Detection

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Abstract—The rise of deepfake technology, driven by advancements in deep learning, poses significant challenges to digital security and privacy due to its potential to create highly realistic yet deceptive media. Detecting deepfakes is crucial to mitigating their misuse. This research analyzes the performance of various deep learning models in detecting deepfakes, focusing on CNN-based architectures. Utilizing the DeepFakeFace dataset, containing 30,000 real and 90,000 AI-generated images, we evaluate the models based on accuracy and other key performance metrics. The study aims to identify the strengths and limitations of these approaches, contributing to the development of reliable deepfake detection systems.

Index Terms—Deepfake detection, Deep learning, Convolutional Neural Networks (CNNs), VGG16, ResNet50, InceptionV3, EfficientNetB3

I. INTRODUCTION

The Rapid developments in machine learning (ML) and artificial intelligence (AI) have drastically changed the digital media landscape. Among the most notable outcomes of these developments is the emergence of deepfakes, a phenomenon wherein synthetic media, such as images and videos, are generated using sophisticated deep learning techniques. The scale of this challenge has reached unprecedented levels, with deepfake files surging from 500,000 in 2023 to a projected 8 million in 2025, representing a staggering 900% annual increase in volume. Deepfake technology leverages powerful algorithms, particularly in the realm of deep neural networks, produce completely fake but incredibly lifelike material. It is frequently challenging to tell the difference between real and modified media because of these artificial inventions, thereby raising critical concerns.

While the innovation behind deepfakes has opened doors to creative applications in entertainment, education, and accessibility, it also presents profound risks. Deepfakes pose serious threats to privacy, digital security, and information integrity, with fraud attempts spiking by 3,000% in 2023 alone. The financial impact is equally alarming, as businesses lost an average of nearly \$500,000 per deepfake-related incident in 2024, with some large enterprises experiencing losses up to \$680,000. The deceptive nature of deepfakes makes them a potent tool for malicious activities, including political propaganda, identity theft, and social engineering attacks. Therefore, the detection and mitigation of deepfake content has become a crucial area of research in the AI community.

Developing robust and reliable deepfake detection systems is essential to combat the misuse of this technology. Recent advances in Convolutional Neural Networks (CNNs) have shown

considerable promise in this area. Recent advances in Convolutional Neural Networks (CNNs) have shown considerable promise in this area, with state-of-the-art models achieving accuracy rates ranging from 87.7% to 99% depending on the architecture and dataset used. CNN-based models are capable of identifying subtle visual artifacts and inconsistencies that often escape the human eye but are indicative of synthetic media. These models can learn to distinguish deepfakes from real content by analyzing spatial features at multiple levels of abstraction.

This research aims to conduct a comprehensive performance analysis of various deep learning models in the context of deepfake detection. Specifically, it will focus on evaluating state-of-the-art CNN architectures in their ability to accurately classify media as real or fake. The urgency of this research is underscored by the fact that 179 deepfake incidents were reported in the first quarter of 2025 alone, surpassing the total number of incidents recorded in all of 2024 by 19%. The study will utilize the DeepFakeFace dataset, a well-known benchmark dataset comprising 30,000 real and 90,000 AI-generated images.

The evaluation will be grounded in critical performance metrics such as accuracy, precision, recall, F1-score, and inference time, offering a holistic view of each model's effectiveness and efficiency. By comparing the performance of different models, this study seeks to highlight both their strengths and limitations, providing actionable insights for researchers and developers working on deepfake detection.

Ultimately, this work aims to contribute to the broader goal of building trustworthy AI systems capable of safeguarding digital content and protecting individuals and institutions from the malicious use of synthetic media.

II. LITERATURE REVIEW

A. Traditional Machine Learning and Handcrafted Feature Approaches

Before the dominance of deep learning methods, several traditional machine learning approaches demonstrated notable effectiveness in deepfake detection tasks. Al-Azawi and Ameen (2023) [2] explored traditional ML classification methods for deepfake image detection, employing Support Vector Machines (SVM) and Random Forest classifiers on handcrafted features, achieving accuracy rates of 78-85% depending on the feature extraction method used. These approaches, while computationally efficient, rely heavily on domain expertise for feature engineering and struggle with the sophisticated

artifacts present in modern deepfake generation techniques. Handcrafted feature approaches have been extensively studied for their interpretability advantages. Xu and Yayilgan (2022) [29] compared SIFT and Histogram of Oriented Gradients (HoG) features with deep learning methods, finding that while handcrafted features achieved 76% accuracy on FaceForensics++ dataset, they exhibited better cross-dataset generalization than some deep learning approaches, maintaining 68% accuracy across different manipulation techniques. Feature point-based methods utilizing FAST keypoint detection and BRIEF descriptors combined with Random Forest classifiers achieved 73% accuracy while requiring significantly less computational resources.

Medical imaging applications have demonstrated the potential of hybrid approaches. Solaiyappan and Wen (2021) [19] evaluated eight different machine learning algorithms including SVM (81% accuracy), Random Forest (84% accuracy), and Decision Tree (79% accuracy) for medical image deepfake detection, comparing them against deep learning models. The study revealed that while traditional methods achieved reasonable performance, they were consistently outperformed by deep learning approaches, with the performance gap ranging from 15-20%.

Despite their computational efficiency and interpretability advantages, traditional methods face significant limitations in handling the increasing sophistication of deepfake generation techniques. Recent studies show that handcrafted feature approaches struggle with high-quality deepfakes, with performance degrading from 85% to 60% when tested on StyleGAN3-generated content.

B. Deepfake Detection with CNN-Based or Deep Learning Architectures

The rapid advancement of generative AI has intensified the need for effective deepfake detection methods. Recent research has explored both proactive defense mechanisms and optimized detection models to address this challenge across various computational environments, revealing significant insights into the evolving landscape of synthetic media countermeasures.

Proactive Defense Mechanisms represent an emerging paradigm that shifts focus from reactive detection to preventive disruption. Wang et al. (2024) [24] introduced "Nullswap," a novel proactive defense mechanism that disrupts deepfake creation by strategically modifying source images to deceive identity extraction algorithms. Unlike traditional reactive detection methods that identify manipulated content post-creation, Nullswap preemptively interferes with the generation process by concealing critical identity features while maintaining visual quality. Their experimental results demonstrate substantial effectiveness, reducing similarity scores between swapped faces to below 0.4 (compared to 0.9 for unprotected images) while preserving image quality with Peak Signal-to-Noise Ratio (PSNR) values exceeding 41 dB. This approach fundamentally transforms the detection paradigm from identi-

fying manipulated content to preventing its accurate creation altogether.

Model Optimization and Efficiency have become critical considerations for practical deployment scenarios. Karathanasis et al. (2025) [9] conducted a comprehensive examination of multiple model compression techniques for efficient deepfake identification in resource-constrained environments. Their knowledge distillation (KD) framework successfully transfers learned representations from a large VGG-based teacher model (4.5 million parameters) to a significantly compressed student model (approximately 10% of the original parameter count). This distillation approach achieved 98% accuracy on benchmark datasets including "140K Real and Fake Faces," consistently outperforming alternative compression techniques such as pruning and quantization. While the effectiveness of knowledge distillation remains dependent on teacher model quality, their findings demonstrate an optimal balance between computational efficiency and detection performance, making real-time deployment feasible.

Generalization Across Generation Methods poses one of the most significant challenges in contemporary deepfake detection research. Wyawahare et al. (2025) [27] conducted an extensive evaluation of CNN architectures across multiple Generative Adversarial Network (GAN) variants, revealing critical insights into cross-domain generalization capabilities. Their initial experiments demonstrated alarmingly low detection accuracies of 20-50% when models encountered deepfakes generated by unfamiliar GAN architectures, highlighting severe generalization limitations that compromise real-world applicability. However, through systematic dataset augmentation incorporating samples from ProGAN, StyleGAN2, and StyleGAN3 architectures, detection performance improved dramatically across all tested models. VGG19 emerged as the most robust architecture, achieving 96-98% accuracy when trained on diversified datasets. Particularly noteworthy is their finding that StyleGAN3-generated deepfakes proved most challenging to detect, achieving the highest visual fidelity and thus representing the current frontier in the ongoing technological arms race between generation and detection methodologies.

III. DATASET AND PREPROCESSING

A. Dataset Collection

The DeepFakeFace dataset represents a meticulously curated collection of facial images specifically assembled for deepfake detection research, sourced from Hugging Face's dataset repository. The dataset comprises 120,000 images in total, with 30,000 authentic images and 90,000 AI-generated deepfake images distributed across multiple generation techniques. The real images component consists of authentic photographs carefully selected from the IMDB-WIKI dataset, which provides a diverse representation of genuine facial imagery across varied demographics, lighting conditions, and photographic styles.

The synthetic image portion is deliberately constructed to encompass multiple contemporary deepfake generation

methodologies: 30,000 images generated via Stable Diffusion V1.5 text-to-image synthesis, exhibiting characteristics of pure AI-generated faces; 30,000 images created through the Stable Diffusion Inpainting model, representing partial facial manipulations where specific facial regions are algorithmically modified; and 30,000 images produced using the InsightFace toolbox, which specializes in face swapping and identity replacement techniques. This diverse composition ensures the model encounters the full spectrum of current deepfake generation approaches. Each image is meticulously labeled with binary classification (real/fake) and contains additional metadata including specific generation method for synthetic images, image resolution, and source information. The inherent class imbalance (25% real vs. 75% fake) reflects real-world detection scenarios where synthetic images typically outnumber authentic ones. We addressed this imbalance during model training by computing class weights using scikit-learn's "compute-class-weight" function with the "balanced" option. This calculation yielded specific weights of 2.00 for class 0 (real images) and 0.67 for class 1 (fake images), precisely accounting for the class distribution in our training set. These weights were then incorporated into the model training process to prevent classification bias toward the predominant fake image class.

B. Data Preprocessing and Augmentation

Our preprocessing pipeline implemented a systematic approach to prepare the DeepFakeFace dataset for optimal model training and evaluation. Initially, we applied Fisher-Yates shuffling with a fixed random seed (42) to ensure reproducibility while randomizing the dataset order. This randomized collection was then partitioned into training (84,000 images), validation (12,000 images), and testing (24,000 images) sets using stratified sampling to maintain consistent class distributions across all subsets.

Image standardization followed a two-step process tailored to each neural architecture's specific input requirements. First, images underwent bilinear interpolation resizing:

- **VGG16 and ResNet50:** 224×224 pixels
- **InceptionV3:** 150×150 pixels
- **EfficientNetB3:** 300×300 pixels

This architecture-specific resizing ensured optimal performance for each model while respecting their original design parameters. Second, pixel intensity normalization converted RGB values from the integer range $[0, 255]$ to floating-point values in $[0, 1]$, preparing them for efficient neural network processing.

To enhance model generalization, we applied targeted augmentation techniques exclusively to the training dataset. These transformations included random horizontal and vertical flipping, alongside brightness and contrast adjustments within a $\pm 10\%$ range. This controlled augmentation strategy effectively expanded the training data distribution without altering validation or test sets, helping models develop invariance to common image variations while significantly reducing overfitting tendencies.

IV. MODELS EVALUATED

A. Resnet50

ResNet50 is a widely-used deep convolutional neural network architecture that is 50 layers deep and is known for its use of residual connections to combat the vanishing gradient problem in very deep networks. It was originally introduced by Microsoft Research and achieved state-of-the-art results on the ImageNet challenge. ResNet50 leverages skip connections that allow gradients to flow more easily during backpropagation, which improves training stability and convergence in deep models.

For the task of deepfake image detection, ResNet50 was employed through a transfer learning approach. The model was adjusted to fit the binary classification job of differentiating between authentic and altered facial photos after being initialised with pretrained weights from ImageNet.

The model architecture was constructed as follows:

- **Base Model:** A pretrained ResNet50 model with ImageNet weights, excluding the original top classification layers (include_top=False). The input shape was set to $224 \times 224 \times 3$, which is standard for ResNet-based models.
- **Fine-Tuning:** The base model was set to be fully trainable (trainable = True) to allow the weights to adapt to the nuances of facial manipulation detection.
- **Global Average Pooling:** A GlobalAveragePooling2D layer was added to reduce the output feature map into a single feature vector per image, maintaining spatial invariance.
- **Fully Connected Layers:**
 - A Dense layer with 256 units and ReLU activation.
 - A Dense layer with 64 units and ReLU activation.
- **Output Layer:** To produce a probability for binary classification (real or fake), a single-unit Dense layer with sigmoid activation was employed.
- **Compilation:** The binary cross-entropy loss function and accuracy were employed as the evaluation metrics, and the model was assembled using the Adam optimizer with a learning rate of 1×10^{-4} . It was trained for 10 epochs with early stopping (patience=3) to restore the best weights based on validation loss. Training used class weights to address imbalance and a ReduceLROnPlateau scheduler for learning rate adjustment.

B. VGG16

VGG16, a well-known convolutional neural network design, was created by the University of Oxford's Visual Geometry Group. It has 16 weighted layers and is renowned for its straightforward, uniform structure using small 3×3 convolution filters and deep depth. VGG16 was originally trained on the ImageNet dataset and became a benchmark in image classification.

For the task of **deepfake image detection**, VGG16 was employed using a **transfer learning** approach. The model was initialized with pretrained weights from ImageNet and

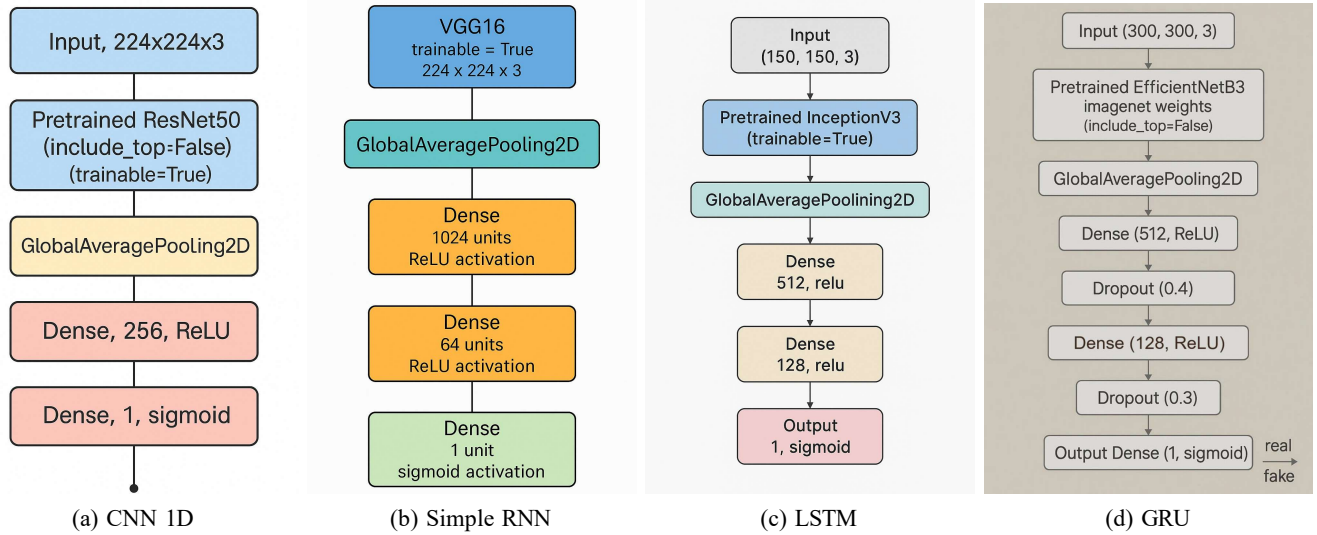


Fig. 1: Architectures of the Models Used

customized for binary classification to distinguish between real and manipulated facial images.

The model architecture was constructed as follows:

- **Base Model:** A pretrained VGG16 model with ImageNet weights, excluding the original top classification layers (`include_top=False`). The standard input shape for VGG-based models was set to $224 \times 224 \times 3$.
- **Fine-Tuning:** The base model was set to be fully trainable (`trainable = True`) to allow all layers to adapt to the domain-specific features of facial manipulation.
- **Global Average Pooling:** A `GlobalAveragePooling2D` layer was added to reduce the output feature maps to a single feature vector per image while maintaining spatial invariance.
- **Fully Connected Layers:**
 - A Dense layer with 1024 units and ReLU activation.
 - A Dense layer with 64 units and ReLU activation.
- **Output Layer:** To produce a probability for binary classification (real or fake), a single-unit Dense layer with sigmoid activation was employed.
- **Compilation:** The binary cross-entropy loss function and accuracy were employed as the evaluation metrics, and the model was assembled using the Adam optimizer with a learning rate of 1×10^{-4} . It was trained for 10 epochs with early stopping (`patience=3`) to restore the best weights based on validation loss. Training used class weights to address imbalance and a `ReduceLROnPlateau` scheduler for learning rate adjustment.

C. InceptionV3

Google created the cutting-edge convolutional neural network architecture known as InceptionV3. It is renowned for making optimal use of computing resources by utilising inception modules and factorised convolutions, which enable the model to learn multi-scale features.

For the task of **deepfake image detection**, InceptionV3 was used via a **transfer learning** approach. The model was initialized with pretrained ImageNet weights and customized for the binary classification problem of identifying real versus manipulated facial images.

The model architecture was constructed as follows:

- **Base Model:** A pretrained InceptionV3 model with ImageNet weights, excluding the original classification layers (`include_top=False`). The input shape was set to $150 \times 150 \times 3$, which is suitable for InceptionV3's architecture and allows reduced computational cost.
- **Fine-Tuning:** The entire base model was set to be trainable (`trainable = True`) so that the network can adapt its features to the deepfake detection domain.
- **Global Average Pooling:** A `GlobalAveragePooling2D` layer was added to convert the convolutional feature maps into a single feature vector per image, improving generalization and reducing overfitting.
- **Fully Connected Layers:**
 - A Dense layer with 512 units and ReLU activation.
 - A Dense layer with 128 units and ReLU activation.
- **Output Layer:** To produce a probability for binary classification (real or fake), a single-unit Dense layer with sigmoid activation was employed.
- **Compilation:** The binary cross-entropy loss function and accuracy were employed as the evaluation metrics, and the model was assembled using the Adam optimizer with a learning rate of 1×10^{-4} . It was trained for 10 epochs with early stopping (`patience=3`) to restore the best weights based on validation loss. Training used class weights to address imbalance and a `ReduceLROnPlateau` scheduler for learning rate adjustment.

D. EfficientNetB3

Google created the EfficientNet family of convolutional neural networks, which uses compound scaling to balance model depth, width, and input resolution. Of them, EfficientNetB3 provides a good balance between efficiency and accuracy, which makes it appropriate for demanding image classification applications. EfficientNetB3, which was pre-trained on the ImageNet dataset, has cutting-edge performance using a notably less number of parameters than conventional architectures.

In the context of **deepfake image detection**, EfficientNetB3 was utilized through a **transfer learning** strategy. The pretrained model was adapted for binary classification to differentiate between real and manipulated facial images.

The model architecture was constructed as follows:

- **Base Model:** A pretrained EfficientNetB3 model with ImageNet weights was used, excluding the original classification layers (`include_top=False`). The input shape was set to $300 \times 300 \times 3$, as required by EfficientNetB3's architecture.
- **Fine-Tuning:** The entire base model was made trainable (`trainable = True`) to allow feature weights to adjust to the specific patterns found in manipulated facial images.
- **Global Average Pooling:** A `GlobalAveragePooling2D` layer was added to compress spatial dimensions, converting feature maps into a single feature vector per image.
- **Normalization and Fully Connected Layers:**
 - A `BatchNormalization` layer to stabilize and accelerate training.
 - A `Dense` layer with 512 units and ReLU activation for high-level feature learning.
 - A `Dropout` layer with a rate of 0.4 to reduce overfitting.
 - A second `Dense` layer with 128 units and ReLU activation for additional abstraction.
 - A second `BatchNormalization` layer followed by a `Dropout` layer with a rate of 0.3.
- **Output Layer:** To produce a probability for binary classification (real or fake), a single-unit `Dense` layer with sigmoid activation was employed.
- **Compilation:** The binary cross-entropy loss function and accuracy were employed as the evaluation metrics, and the model was assembled using the Adam optimizer with a learning rate of 1×10^{-4} . It was trained for 10 epochs with early stopping (`patience=3`) to restore the best weights based on validation loss. Training used class weights to address imbalance and a `ReduceLROnPlateau` scheduler for learning rate adjustment.

V. RESULTS

To assess the effectiveness and generalization capability of various CNN-based transfer learning models, a series of controlled experiments were conducted. Each experiment followed a consistent pipeline, ensuring that all models were trained

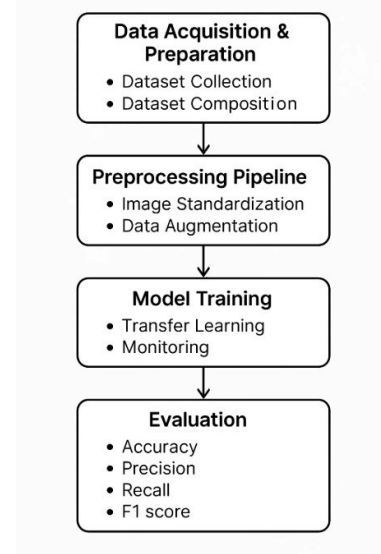


Fig. 2: **Methodology Pipeline.** Overview of data preprocessing, model training, and evaluation process.

and tested on the same preprocessed dataset, under identical training conditions. This approach helped maintain fairness in evaluation and minimized the influence of external variables such as data distribution shifts or training inconsistencies.

The models selected for comparison: ResNet50, VGG-16, InceptionV3, and EfficientNetB3, are among the most widely adopted convolutional neural network architectures, known for their performance on image classification tasks. These models were initialized with ImageNet weights and fine-tuned on the target dataset to adapt them to the specific classification objective.

For performance evaluation, we used four key classification metrics: accuracy, precision, recall, and F1-score. These metrics provide a holistic view of the model's performance by measuring not only how often predictions are correct (accuracy), but also how well the model distinguishes between classes, particularly in the presence of class imbalance (precision, recall, and F1-score). By analyzing all four metrics, we ensured a comprehensive understanding of each model's strengths and limitations in predicting the correct class labels.

Models Evaluated	Accuracy	Precision	Recall	F1-score
ResNet50	0.98	0.98	0.98	0.98
VGG-16	0.92	0.94	0.91	0.91
InceptionV3	0.88	0.90	0.86	0.88
EfficientNetB3	0.76	0.76	0.76	0.76

TABLE I: Performance comparison of models on classification task. All metrics are reported in the range of [0, 1].

The experimental results demonstrate a clear hierarchy in performance among the tested models for deepfake detection, with ResNet50 emerging as the superior architecture. Key findings include:

ResNet50: Exceptional Performance: ResNet50 demonstrated outstanding performance, achieving a consistent score of 0.98 across all evaluation metrics. This uniformity suggests that the model is exceptionally well-balanced, minimizing both false positives (authentic images misclassified as deepfakes) and false negatives (deepfakes misclassified as authentic). The high recall value of 0.98 is especially important, as it indicates the model’s ability to identify nearly all deepfake images, a crucial trait for real-world deployment. The model’s residual learning framework, which leverages deep residual connections, allows it to capture subtle visual artifacts and inconsistencies characteristic of deepfake images. These features, which may otherwise be overlooked, are key to its ability to consistently detect deepfakes with high accuracy.

VGG-16: Robust, but Slightly Less Effective: VGG-16 ranked second in the experiments, achieving an accuracy of 0.92. It performed with slightly higher precision (0.94) compared to recall (0.91), indicating that while it effectively reduces false positives, it occasionally misses some deepfake images. The sequential convolutional layers of VGG-16 allow it to capture fine-grained features, which help in recognizing deepfake artifacts. However, its simpler architecture compared to ResNet50 means that it lacks the deeper, more complex residual learning capability, leading to slightly lower recall. While still a strong contender, VGG-16’s performance shows that its ability to capture high-level features is robust but not as fine-tuned as ResNet50’s.

InceptionV3: Moderate Performance: InceptionV3 showed moderate performance, with an accuracy of 0.88. While its precision was reasonable (0.90), its recall of 0.86 was noticeably lower. This lower recall is a significant concern, as it indicates that the model misses a higher proportion of deepfake images compared to the top two models. InceptionV3 uses a multi-scale processing approach, allowing it to detect features at different resolutions. However, this approach did not appear to be as effective at identifying the subtle visual inconsistencies found in deepfake images. The model’s structure may excel in capturing a broad range of features, but the specific artifacts of deepfakes seem to require more specialized detection capabilities that ResNet50 offers.

EfficientNetB3: Struggled to Capture Deepfake Artifacts: Despite its innovative compound scaling approach that aims to balance network depth, width, and resolution, EfficientNetB3’s comparatively low performance (0.76 across accuracy, precision, recall, and F1) appears to stem from an architectural mismatch with forensic cues central to deepfake detection. Its compound scaling and depthwise separable convolutions prioritize efficiency by aggressively balancing depth, width, and input resolution, but this can attenuate high-frequency residuals, compression artifacts, and subtle blending inconsistencies that often distinguish synthetic from authentic faces. Pretraining on ImageNet further biases the feature space toward semantic content rather than artifact-centric signals, limiting transfer effectiveness on manipulation forensics relative to architectures with strong residual pathways (e.g., ResNet50), which better preserve fine-grained detail

across layers. Another factor is capacity: while EfficientNetB3 is parameter-efficient, the reduced representational budget can constrain learning of diverse artifact distributions (e.g., generator-specific noise patterns, upsampling footprints), especially under class imbalance and heterogeneous generation pipelines. In short, without domain-specific adaptations, such as frequency-domain augmentation, anti-aliasing strategies, or ensembles combining EfficientNet with artifact-focused backbones, compound scaling’s efficiency gains come at the cost of diminished sensitivity to the subtle, high-frequency irregularities that characterize modern deepfakes, explaining the observed gap to ResNet50 and the smaller margin to VGG-16 in our experiments.

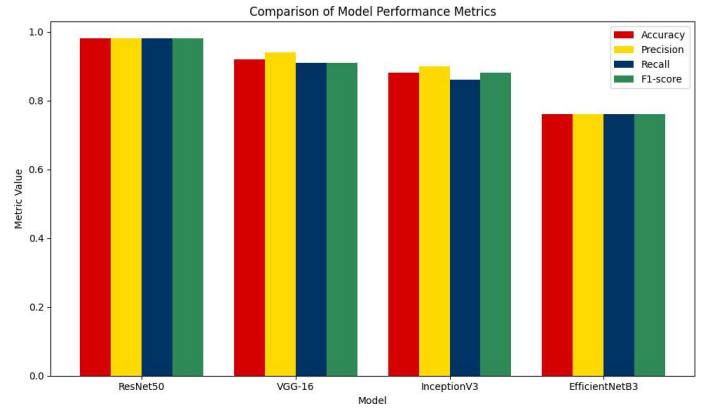


Fig. 3: **Comparison of Accuracy, Precision, Recall, and F1-score** for ResNet50, VGG-16, InceptionV3, and EfficientNetB3.

VI. CONCLUSION

This research compared four CNN architectures: ResNet50, VGG-16, InceptionV3, and EfficientNetB3, on the DeepFake-Face dataset to evaluate their suitability for deepfake detection. ResNet50 achieved the highest performance (0.98 on accuracy, precision, recall, and F1-score) thanks to its deep residual learning framework, which excels at capturing subtle visual inconsistencies inherent to deepfake media. VGG-16 followed with strong but slightly imbalanced performance (0.92 accuracy, higher precision than recall), reflecting its capability to learn fine-grained features but occasional missed detections. InceptionV3 (0.88 accuracy) and EfficientNetB3 (0.76 accuracy) underperformed, underscoring that architectures optimized for general image classification may not directly translate to specialized forensic tasks without targeted adaptations.

Beyond academic benchmarks, these findings have important real-world implications. In digital forensics and law enforcement, high-recall models like ResNet50 can be deployed to scan vast archives of images and videos to flag manipulated content, aiding investigations into identity theft, child exploitation, and terrorist propaganda. Similarly, social media platforms and content-moderation services can integrate these models into automated pipelines to detect and remove

deepfake posts in real time, mitigating misinformation before it spreads to millions of users. In financial services, where deepfake-powered phishing and impersonation attacks have surged by over 3,000%, deploying robust detection at customer touchpoints (video KYC, remote onboarding) can prevent fraudulent transactions and reduce annual losses averaging \$500K per incident. Moreover, journalism and media outlets can use deepfake detectors to verify authenticity of user-generated content during breaking news coverage, preserving public trust and editorial integrity.

While EfficientNetB3 offers computational efficiency suited to edge devices and mobile applications, its lower accuracy suggests a hybrid approach: combining lightweight models for rapid screening with heavyweight architectures like ResNet50 for detailed analysis. Future work should extend evaluations to video-based deepfakes, explore model compression via knowledge distillation to balance performance and resource use, and integrate proactive defenses (e.g., Nullswap) alongside reactive detection to both prevent and identify manipulated media. Enhancing model explainability will further build user and stakeholder trust, paving the way for widespread adoption of deepfake detection systems in critical security, media, and commercial environments.

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